WASTEWISE SCAN · PROJECT PLAN

WasteWise Scan project plan

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First project plan with resources, stage gates and references (kit 1.8 upgrade)

WasteWise Scan project plan

WasteWise Scan is at stage P3 (design outputs, TRL 3): option A is modelled at D3, drawn, priced and planned to build, and option B is modelled at D3 for comparison. The next gate, P4 (verification, TRL 4), is on hold under the portfolio cap. The decisions requested from Amish are the concept selection between A and B (within 10 %, no pick) and the extended InGaAs band (WSC-DEC-001 open decisions 1 and 2). Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Optics and calculations	Design lead	WSC-CAL-001, WSC-STM-001	Done at TRL 3; fix F1 in
Mechanical design and CAD	Design lead	`cad/src/model.py` (D3, both options), WSC-DWG-001, DWG-101 to 109	Done at TRL 3
Build plan	Design lead	WSC-BLD-001	Done (plan, not built)
Requirements and traceability	Design lead	WSC-URS-001, WSC-PRS-001, WSC-RTM-001	Done; targets to confirm in co-design
Co-design and partner	Project lead (Amish)	Sessions with a waste picker cooperative (first candidate SWaCH, Pune), price grades, labelled items	Not started (outreach is Amish's)
Shared record with WasteWise-ml	Project lead	WSC-DDR-002 adopted on both sides	At risk until WasteWise-ml adopts it
Regulatory	Procurement with a qualified reviewer	WSC-REG-001 items to assess (cell, radio, eye safety)	Register written; items to assess
Verification (P4)	Test lead	Test plans and lab reports (TST)	On hold (TRL cap 3)

Resources needed

Table 2. Resources. Costs are estimates for planning, not quotes.

Resource	Deliverable	Effort	Estimated cost (USD)
Parts for one prototype	Scanner, cap, spares of the printed parts	Purchase	167.50 (BOM); up to about 260 with a branded photodiode (WSC-MKT-001)
Labelled item set	200 or more clean items of known resin, collected with the partner	2 to 3 days with the partner	To assess
Maker time	Build to WSC-BLD-001	About 3 days	To assess
Co-design facilitation	Two sessions with the cooperative	2 days	To assess
Regulatory review	Cell, radio and eye-safety items of WSC-REG-001	Half a day	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB, STM, OPT, REG, FEA, PLN	Met (2026-10-24)
P1 Definition	URS, PRS draft, RSK with hazard analysis, MKT	Met (2026-10-24)
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR, STEM closed, feasibility	Met except the STEM check stays open (S22, S23 need users and items) and the concept selection awaits Amish
P3 Design outputs	Model, STEP, drawings, priced BOM, build plan, DEC, design FMEA, traceability	Met (2026-10-24)
P4 Verification	Lab tests of optics, accuracy on labelled items, drop and spray, timing and battery	On hold (TRL cap 3)
P5 Validation	Field trial with the cooperative; human factors	On hold
P6 Commercialization	Only with a named deployment partner	On hold

Next steps and decision requested

1. Amish decides the concept selection (WSC-DEC-001 open decision 2): A, the current design, or B; no pick because the scores are within 10 %.
2. Amish decides open decision 1 (extended InGaAs band now, or after the first labelled item set).
3. Approach SWaCH in Pune jointly with WasteWise-ml (decided 2026-10-02; outreach is Amish's).
4. When TRL 4 is opened: build to WSC-BLD-001, measure LED spectra, the photodiode curve, window crosstalk with fix F1, cap opacity and tube opacity, then scan the labelled item set.

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